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Isolation and identification of bacteria and actinomycetes from work place environment of sugar industry.

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Abstract : The sugar industry is the largest agro based industry plays a key role in economy of Maharashtra state. The occupational health problems in workers working in various processing units of sugar industry are enormous, mainly because of occupational stress factors, viz. high intensity noise, temperature, high concentration of dust,inadequate illumination, inadequate space, shift work and occupational diseases due to microbial inflection .In present investigation infectious bacteria and actionomycetes were detected from five different processing units of sugar industry during working period .The predominantly bacterial species observed were primarily identified as *Staphylococcus spp., Micrococcus spp., Flavobacterium spp*. However ,Actinomycetes a spp. of *thermoactinomycetes* primarily identified as *Thermoactinomycetes sacchari* was isolated.

Key Words: Sugar Industry, Actinomycetes ,Occupational Health.

INTRODUCTION

Sugar industry constitutes one of the largest sector of agrobased industries in our county and occupies very important position in the economic development of the country. The sugar industries constitution to national economy is substantial, sugar factories located in various parts of the country act as nuclei for the development of the rural areas by mobilizing rural resources, generating employment transport and communication facilities over 45 million farmers, their dependents and a large mass of agriculture labour (7.5% of the rural population in all) are involved in the sugarcane cultivation ,harvesting and ancillary activities up to sugar production .

Although sugar industry is largest industry it is running without research or innovation of any kind. Not much attention has been paid to the safety and health in sugar industry. There are many occupational health problems in this sector, which are attributable to a variety of occupational stresses such as excessive heat, inadequate light, high intensity noise as well as excessive work load and unsatisfactory working conditions. Higher concentration of the dust particles are mainly found in cane yard, mill, boiler, bagasse baling and sugar house processing units and the workers exposed to dust are affected by various problems of repository system.

The dust is generated in the atmosphere as a result of transportation of the sugarcane and its processing. Bagasse is the first byproduct formed after extraction of juice from sugarcane .Bagasse consists of cellulose, hemicelluloses, ash, sulphur, potassium and 40 to 49 percent of moisture.

Bacteria and fungi associated with bagasse dust are potential source of water soluble compound that can cause symptoms similar to acute response of bagassosis in susceptible workers so the bagasse dust is complex heterogeneous material occur in its biological content and activity .Bagassosis occurs when sugar mill workers inhale the dust from bagasse (Qmar Rahman *et al*.,2001). Schuyer(2002) studied that the thermophilic actinomycetes are ubiquitous organism usually found in bagasse and bagassosis result from thermophilic actinomycetes.

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MATREIALS AND METHODS

Study Area

The present study was carried out at Hutatma kisan Ahir Sugar industry having 4500 tons crushing capacity per 24 hours. Walwa is the tehsil place of the sangli district, Maharashtra state.

Microbial Examination

Representative air samples from the areas of the following various processing units (sections)of the sugar industry were selected for microbial examination:

A) Cane Yard Section B)Engineering Section – It Constitutes a) Mill-i) Entrance of the mill ii)Inside the mill b) Boiler and c) Bagasse baling

1. Isolation and Enumeration of the air Microorganisms:

Gravity sedimentation method [The gravity petridish (settle plates technique)] as described by Gregory (1973) in the microbiology of the atmosphere and Chruiteshank et al.(eds)(1975) in "Medical Microbiology " was used for isolation and enumeration of the microorganism.

A) Isolation ,Enumeration and identification of Bacteria

I. Isolation and enumeration of bacteria:

For isolation and enumeration of the bacteria appropriate number of petri dishes containing sterile nutrient media (spc and blood agar) were exposed to the air in the various above referred processing units of the sugar industry for various period viz.,0.5 min.,1 min,.1.5 min., 2 min. and 3 min. and then one set of plates of each medium was incubated at room temperature while other set of plates was incubated at 37° C for 24 to 48 hours .Then colony count was taken ,plates having colonies 30 to 300 were selected for counting. Then well isolated representative colonies were picked up and transferred on the fresh sterile media and subjected to purification.

II. Identification of Bacteria:

Predominant bacterial isolates were studied for morphological, cultural and physiological characteristics and then primary identified with reference to Bergey's manual of systematic bacteriology Vol.I,II,III and IV(1984,86 and 1989) .

a) Morphological Characteristics:

Morphological characteristics were mainly studied by Gram's reaction (Huchers modification).

b) Cultural Characteristics:

In the studies on the cultural characteristics, motility was studied by hanging drop preparation and colony characteristics of the isolates were studied by noting the shape, size, consistency, elevation etc. of well isolated colonies.

c) Physiological and Biochemical Characteristics:

In the studies on the physiological and biochemical characteristics of bacterial isolates following test as detailed by Chruiteshank et al. (eds.) 1975,except otherwise stated were performed viz.Oxidase test, Catalase test, Benzidine test(Skerman 1967), Growth and pigmentation on nutrient agar, Methyl red test, Voges-proskauer test , Indole production test,Citrate (Simmon's) utilization test ,Nitrate reduction test ,Hydrogen sulfide production test , Litmus milk test ,Urease test(Christenson's),Lecithinase test(Egg yolk reaction),Acid gas and acetoin production from glucose .In case of isolates belonging to genus Bacillus [Buchanan et al.(eds.),1974], Anaerobic growth by stab culture technique in deep agar tubes**,** Growth and acid production aerobically and anaerobically from glucose and mannitol, Phenylalanine deamination test ,Esculine hydrolysis test, Growth on centrimide test, Fermentation of carbohydrates viz. D glucose,D-lactose,D-xylose, Dmaninitol, Larabinose, Dulcitol,sucrose, Esculin, Inositol,

Maltose, Salicin,Sarbitol, trehalose,Amino acid decarboxylation test, Tyrosine decomposition test(Collins *et al*,1975), Gelatineliauefaction test, ,Growth in presence of various concentration of NaCl,Growth in Sabouraud's dextrose broth, Case in hydrolysis test(in milk agar), Growth in presence of 0.001% lysozyme ,Starch hydrolysis test,Arginine hydrolysis test, Growth in presence of 0.02% sodium azide,Hippurate hydrolysis test ,Phosphates test Growth at different temperature ,Growth at Various pH,Growth on MacConkey's agar,

B.Isolation, Enumeration and Identification of Actinomycetes

I. The Isolation and Enumeration

For isolation of actinomycete petrisidhes containing Glycerol Aspargine agar were exposed to the air in the various above referred sections for various periods viz.0.5

min.,1min.,1.5 min.,2 min., and 3 min. then plates were incubated to room temperature for 2 to 5 days and one additional set of plate was incubated at 55°C. the actinomycetal count was taken, plates having colonies below 50 to 60 were selected for actual count to find out actinomycete population.Representative well isolated colonies were transferred into fresh medium for purification by isolation and after 48 hours colony characteristics of actinomycet were noted and actinomycetal isolates were subjected to microscopic examination .

II. Identification of Actinomycetes:

Primary identification of predominant actinomycetes upto genus level was done from morphological and cultural characteristics of the isolates.

RESULTS

Table No.1.shows bacterial colony count obtained on nutrient agar media.It can be seen from the table that cane yard air environment shows uncountable bacterial colonies at all exposure periods viz. 0.5 min.,1.0 min.,2 min. and 3 min indicating high bacterial number in the air environment of the cane yard .This could be mainly because of high density of the dust particle which in turn could be because of immense human and animal activities in the cane yard.

Table 1:Rresults of the bacteriological examination of the air (bacterial count)

Sr. No.	Sections of the sugar industry examined	Colony count on nutrient agar after incubation at room temperature for 24 hours					Colony count on nutrient agar after incubation at 37^0C temperature for 24 hours				
		Period of exposure of plates in the air environment in minutes					Period of exposure of plates in the air environment in minutes				
		0.5	1.0	1.5	2.0	3.0	0.5	1.0	1.5	2.0	3.0
1	Cane yard Section	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable	Uncountable
2	Entrance of Mill section	260	307	Uncountable	Uncountable	Uncountable	90	131	140	232	268
3	Inside the mill section	142	224	305	Uncountable	Uncountable	114	187	247	Uncountable	Uncountable
4	Boiler Section	32	68	92	113	182	22	57	88	102	134
5	Bagasse baling Section	79	134	142	156	173	72	112	137	160	187

Table 2 :results of isolation and identification of bacteria and actinomycetes

Sr.NO.	Isolate	Gram Nature	Genera/Species
1	C-1	+ve	***Staphylococcus ayrcus***
2	C-2	+ve	***S.epidernidis***
3	C-3	-ve	***Serratiamarscences***
4	C-4	- ve	***Pseudomononas***
5	C-5	+ve	***Micrococcus roseus***
6	C-6	+ ve	***Bacillus subtilis***
7	ME-1	- ve	***Flavobactrium***
8	ME-2	+ve	***Bacillus***
9	ME-3	+ve	***Bacillus***
10	ME-4	+ ve	***Micrococcus luteus***
11	ME-5	-ve	***Arthrobacter spp.***
12	MI-1		***S.albus***
13	MI-2	+ ve	***B.subtilis***
14	MI-3	+ ve	***B.cereus***
15	MI-4	- ve	***Pseudomas spp.***
16	MI-5	-ve	***Aeromonas spp.***
17	B-1		***Micrococcus rosen***
18	B-2	+ve	***Bacillussubtilis***
19	B-3	+ve	***S.albus***
20	B-4	-ve	***Serratiamarscence***
21	B-5		***Thermoactinnomyces spp.***
22	Bag-1	+ve	***Bacillus cereus***
23	Bag-2	+ve	***Bacillus cereus***
24	Bag-3	+ve	***Pseudomonas spp.***
25	Bag-4	+ve	***Micrococcus luteus***
26	Bag-5		***Thermoactinomyces spp.***

C=Cane Yard ME=Entrance of mill section
MI=Inside the mill section B=Boiler Section Bag=Bagasse baling section

Some of these species belonging to the category of isolates are known to be pathogenic for man, for instance certain species of Pseudomonas, Staphylococci and Bacillus.

Table No.3 shows the actinomycetal colony count obtained on glycerol aspargin agar. It can be seen from the table that the air environment of cane yard section shows 10 actinomycetal colonies ,the air environment at the entrance of the mill and inside mill shows 7 and 9 actinomycetal colonies with 1.5 min exposure period respectively .It is further seen that air environment of boiler section has 8 actinomycetal colonies with 1.5 min exposure period. It could also be seen from Table No.3 that air environment of bagasse baling section has considerable actinomycetal number viz.15 colonies with 1.5 min. exposure period.

It is interesting to note that as indicated in the Table No. 2 that Actinomycetes a spp. Of thermoactinomycetes primarily identified as *Thermactinomycetes sacchari* was isolated from the air environment of boiler and bagasse baling section .The detection of this organism is very significant because the inhalation of spores of these organisms can lead to interstitial lung disease in workers handling bagasse known as bagassossis.

Table 3: Result of the colony count of the actinomycetes on glycerol aspargin agar at room temperature

Sr.No.	Sections of the industry examined	Period of the exposure of the plates to the air environment in minutes				
		0.5	1.0	1.5	2.0	3.0
1.	Cane Yard	3	8	10	15	21
2.	Entrance of Mill section	2	4	7	9	13
3.	Inside the mill section	1	6	9	12	18
4.	Boiler Section	3	5	8	11	15
5.	Bagasse baling Section	4	7	15	19	28

DISCUSSION

Agarwal and Agarwal (1994) suggested that the exposure to dust produces pulmonary edema, bronchial asthma, pulmonary mycosis, bronchopulmonary aspergillosis, rhinitis, sinusitis, otitis, irritability and inflammation of the heart.

In the present investigation, it has been demonstrated that the concentration of respirable dust particles is significantly high at cane yard and bagasse baling section.In cane yard section due to vehicle transportation and unloading of sugarcane, the concentration of clay dust is very high. In bagasse baling section bagasse dust formed after the extraction of juice from sugarcane is found in high concentration. These respirable dust particles easily reach the wall of alveoli and may cause inflammation leading to the respiratory impairments. Air environment of cane yard entrance of mill,inside the mill,boiler and bagasse baling section is contaminated with large number of Gram negative and Gram positive bacteria.The endotoxins of Gram negative bacteria are responsible for several toxic effects have been studied by Cinkotoi(1977).

The most predominant bacterial species identified in the present investigation were belonging to genus, S*taphylococcus ,Serratia, Bacillus, Pseudomonas, Micrococcus and Flavobactrium.* It is also known that certain species of pseudomonas , staphylococci and bacillus are pathogenic for human,Besides the bacteria air environment of cane yard, entrance of mill, inside the mill, boiler and bagasse sections of sugar industry is contaminated with actinomycetes.

It has been proved that occupational disease bagassosis in worker handling mouldy sugar cane residue is caused by related thermophilic actionomycetes (Larcey,1971) .Biggins and Abrahamas (1991) documented the bagassosis is a recognised occupational lung disease of the extrinsic allergic alveolits type caused by spores of thermophilic actionomycetes. Singh *et al*(1999) reported that in the bagasse unit and cane cutting sections of the sugar refinery thermophilic fungi and actinomycetes were important component of the air. Schuyer(2002) studied that the thermophilic actinomycetes are ubiquitous organisms usually found in the bagasse and bagassosis results from thermophilic actinomycetes i.e.*Thermoactinomycete ssacchari ,T.Vulgaris*.

In our finding a species of thermoactinomycetes primarily identified as *Themoactionmyce saccharii* was isolated from the air environment of boiler section and bagasse sections.

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